Wireless Protocols for IoT : Zigbee

1. Zigbee Features, Versions, Device Types, Topologies
2. Zigbee Protocol Architecture
3. Zigbee Application, Zigbee Application Support Layer
4. Network Layer, Routing: AODV, DSR
5. Zigbee Smart Energy V2

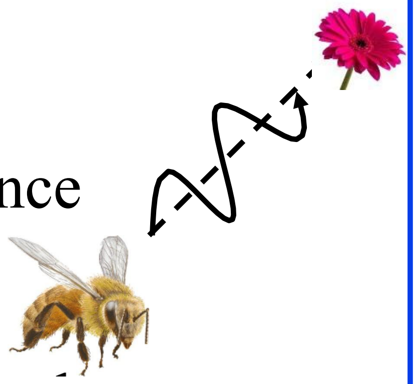
Note: Bluetooth, IEEE 802.15.4 were covered in the previous lectures..

Zigbee Overview

- Industrial monitoring and control applications requiring small amounts of data, turned off most of the time (<1% duty cycle), e.g., wireless light switches, meter reading, patient monitoring
- First standard was published in 2004
- Ultra-low power, low-data rate, multi-year battery life
- Power management to ensure low power consumption.
- Less Complex. 32kB protocol stack vs 250kB for Bluetooth
- **Range**: 1 to 100 m, up to 65000 nodes.
- **Tri-Band**:
  - 16 Channels at 250 kbps in 2.4GHz ISM
  - 10 Channels at 40 kb/s in 915 MHz ISM band (USA)
  - One Channel at 20 kb/s in European 868 MHz band
  - 920 MHz in Japan

Zigbee Overview (Cont)

- IEEE 802.15.4 MAC and PHY
  (Except for Zigbee Smart Energy 2.0)
  Higher layer and interoperability by Zigbee Alliance
- Up to 254 devices or <u>64516</u> (~2^{16}) simpler nodes



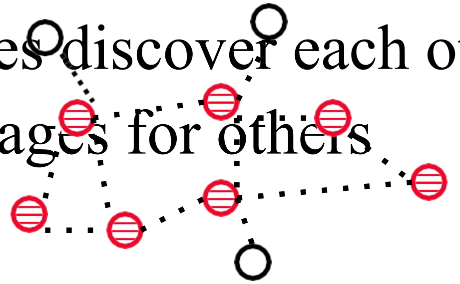
- Named after zigzag dance of the honeybees
  Direction of the dance indicates the location of food
- Multi-hop ad-hoc mesh network

Multi-Hop Routing: message to non-adjacent nodes

Ad-hoc Topology: No fixed topology. Nodes discover each other

Mesh Routing: End-nodes help route messages for others

Mesh Topology: Loops possible

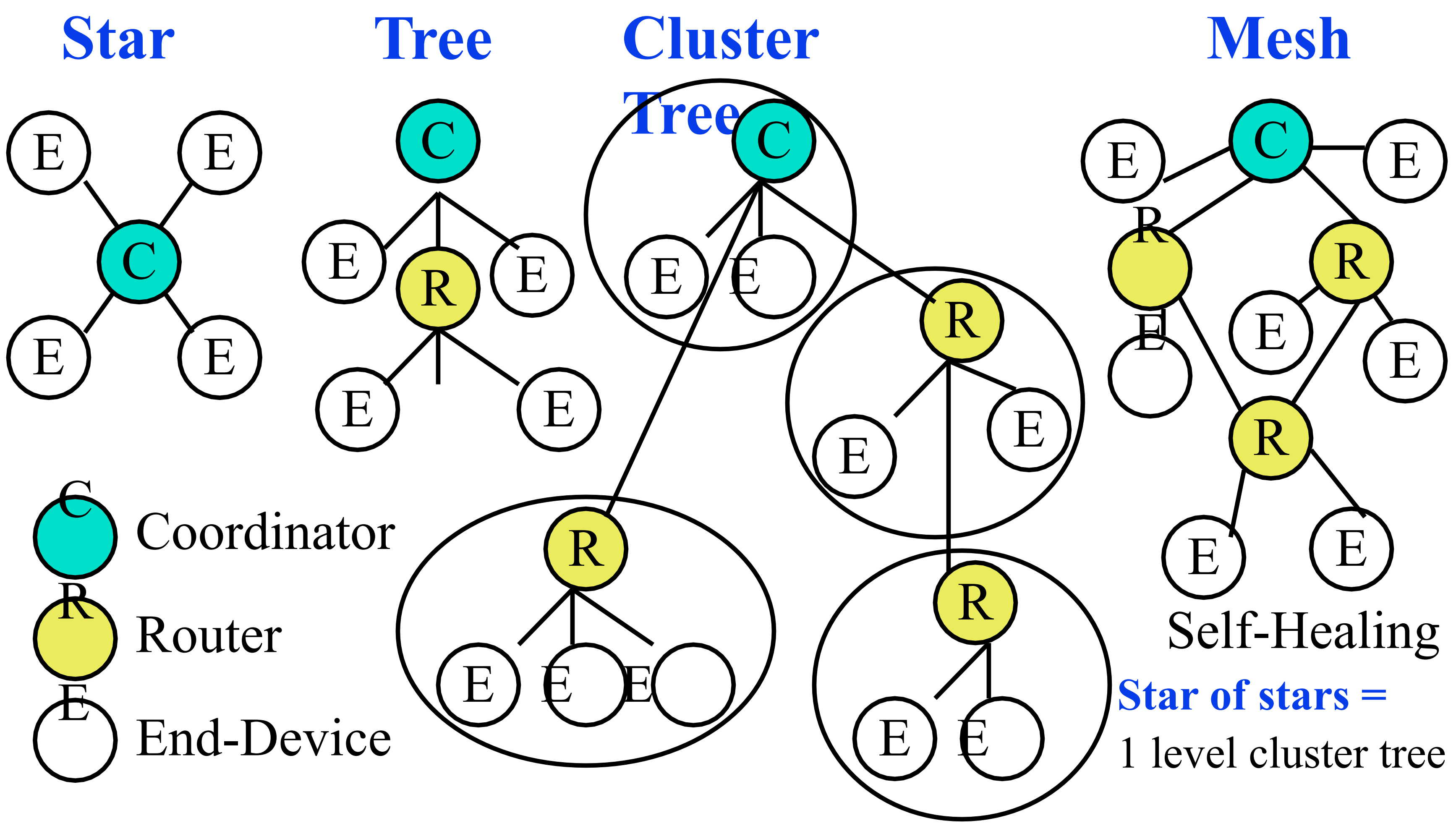


Ref: Zigbee Alliance, http://www.Zigbee.org

Zigbee Device Types

- **Coordinator**: Selects channel, starts the network, assigns short addresses to other nodes, transfers packets to/from other nodes
- **Router**: Transfers packets to/from other nodes
- **Full-Function Device**: Capable of being coordinator or router
- **Reduced-Function Device**: Not capable of being a coordinator or a router $\Rightarrow$ Leaf node
- **Zigbee Trust Center (ZTC):** Provides security keys and authentication
- **Zigbee Gateway**: Connects to other networks, e.g., WiFi

Zigbee Topologies



Example Smart Home Devices

Device	Zigbee Role	Function
Smart Home Hub (e.g., Echo Plus, SmartThings)	**Coordinator (ZC)**	Manages network, processes commands
Smart Light Bulbs (e.g., Philips Hue)	**Router (ZR)**	Controls lighting, relays signals
Smart Plugs	**Router (ZR)**	Turns appliances on/off, extends Zigbee range
Motion Sensor	**End Device (ZED)**	Detects movement, triggers automation
Smart Door Lock	**End Device (ZED)**	Provides remote locking/unlocking
Smart Thermostat	**Router (ZR)**	Controls heating and cooling

Scenario: Automated Home Lighting & Security System
Imagine you have a **Zigbee-enabled smart home** where lights, locks, and sensors work together. The system is programmed to:
Turn on the lights when someone enters a room.
Lock the door automatically at night.
Adjust the thermostat based on room occupancy.

Step 1: Zigbee Network Setup
The **Zigbee Coordinator (Smart Hub)** creates a **Personal Area Network (PAN)** and assigns a **PAN ID**.
Zigbee devices join the network using the **beacon-enabled IEEE 802.15.4 MAC protocol**.

Smart Hub (ZC): "Creating Zigbee Network - PAN ID 0x1234"
Smart Light (ZR): "Joining Zigbee Network with PAN ID 0x1234"
Motion Sensor (ZED): "Connected to PAN ID 0x1234"

Step 2: Motion Detected – Lights Turn On
A **Zigbee motion sensor** detects movement in the living room.
The motion sensor **sends a Zigbee message** to the **Smart Hub**.
The **Smart Hub sends a command** to the **Zigbee light bulb** to turn on.

Motion Sensor (ZED) → Smart Hub (ZC): "Motion detected in living room!"
Smart Hub (ZC) → Smart Light (ZR): "Turn ON!"
Smart Light (ZR): "Light is ON"

Step 3: Smart Lock Engages at Night
At **10:00 PM**, the **Smart Hub sends a command** to the **Zigbee Smart Lock**.
The door **automatically locks for security**.

Smart Hub (ZC) → Smart Lock (ZED): "Time: 10:00 PM - Lock the door!"
Smart Lock (ZED): "Door Locked"

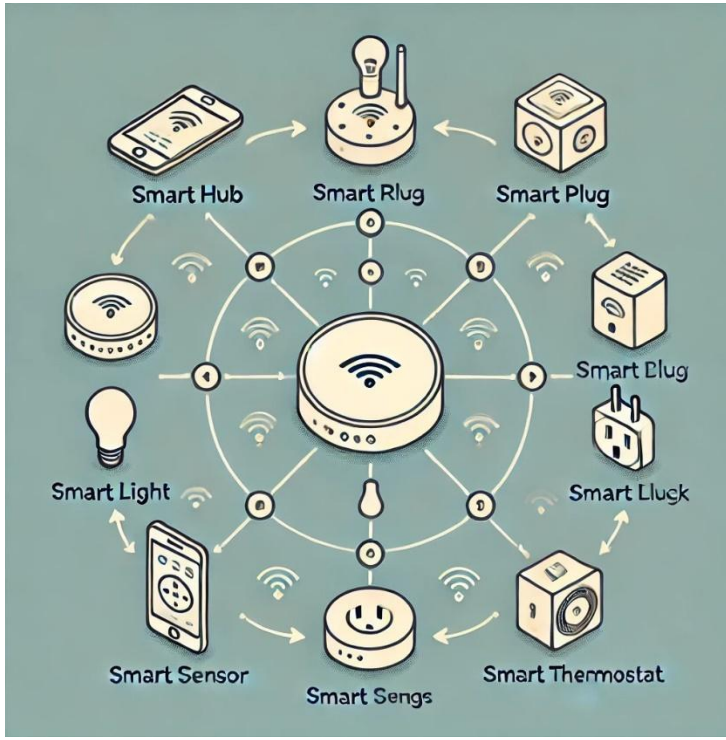
Step 4: Adjusting the Thermostat Based on Occupancy
If **no motion is detected for 30 minutes**, the **Smart Thermostat** lowers the temperature.
The **Zigbee thermostat communicates** with the Smart Hub.

Motion Sensor (ZED) → Smart Hub (ZC): "No motion detected for 30 minutes"
Smart Hub (ZC) → Smart Thermostat (ZR): "Set temperature to 20°C"
Smart Thermostat (ZR): "Temperature adjusted to 20°C"

Step 5: Remote Control via Mobile App

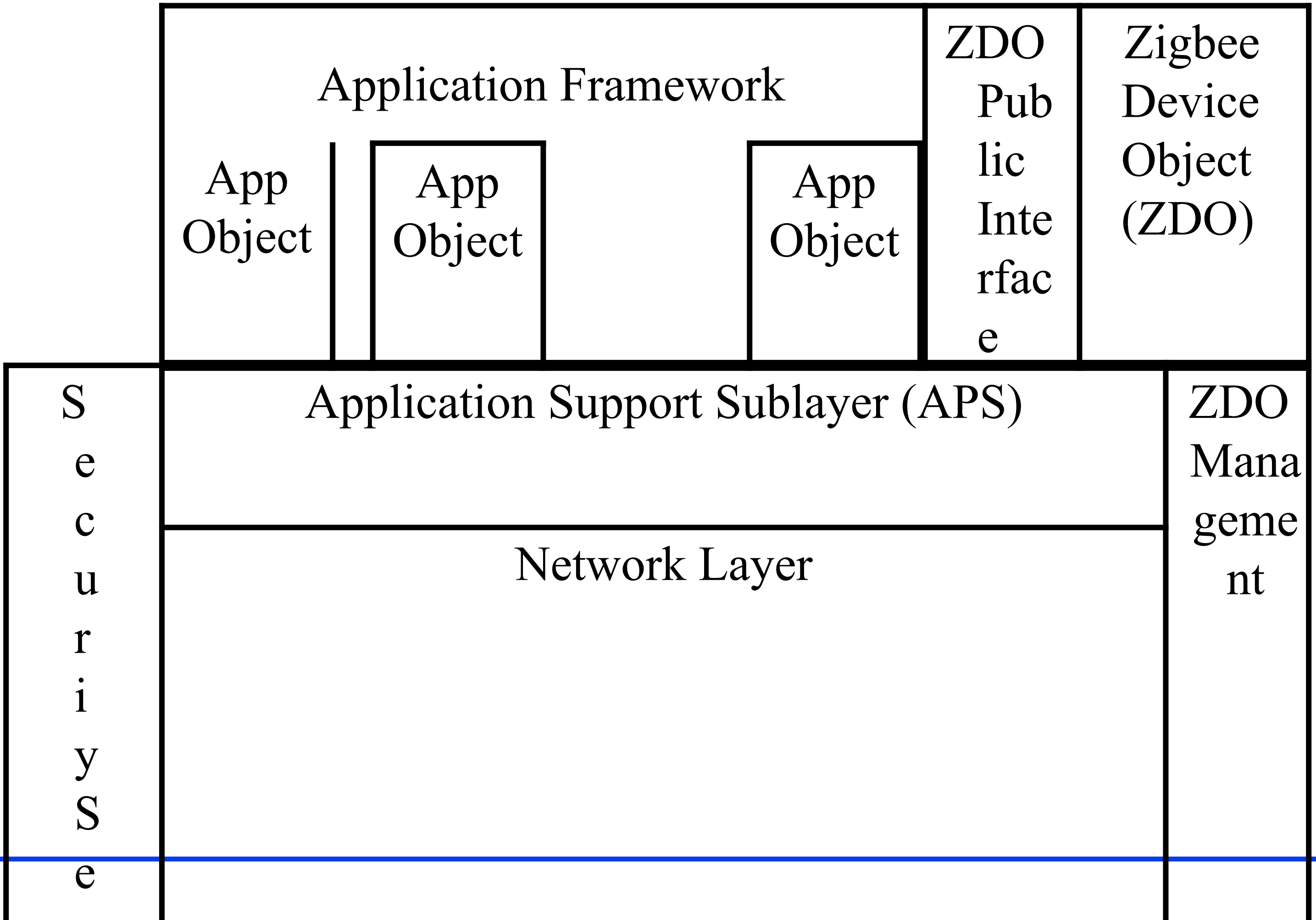
The user **opens a Zigbee-compatible mobile app** (e.g., Amazon Alexa, SmartThings).
The app **sends a Zigbee command** via the **Smart Hub** to turn off all devices.

User (App) → Smart Hub (ZC): "Turn off all lights and devices"
Smart Hub (ZC) → All Zigbee Devices: "Power OFF"
Smart Light (ZR): "Light is OFF"
Smart Plug (ZR): "Appliance is OFF"



- **AI-Based Automation** – Predictive control based on user behavior.
- **Voice Control with AI Assistants** – Seamless integration with Alexa, Google Assistant.
- **Energy Optimization** – AI-powered energy-saving algorithms.
- **Integration with Matter (IoT Standard)** – Cross-device compatibility.

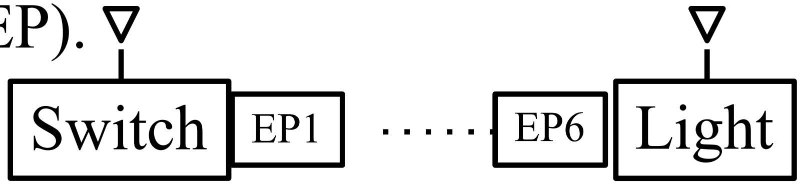
Zigbee Protocol Architecture



Layer	Function	Key Technologies
Application (APL)	Device roles, commands, user applications	APS, ZDO
Network (NWK)	Routing, addressing, network formation	Mesh, AODV
MAC	Medium access, beaconing, collision avoidance	CSMA/CA, ACK
Physical (PHY)	Modulation, transmission, reception	IEEE 802.15.4

Zigbee Protocol Architecture (Cont)

- **Application Objects**: e.g., Remote control application. Also referred to as **End-Point** (EP).



- **End-Node**: End device. Each node can have up to 250 application objects.
- **Zigbee Device Object (ZDO)**: Control and management of application objects. Initializes coordinator, security service, device and service discovery
- **Application Support Layer (APS)**: Serves application objects.
- **Network Layer**: Route Discovery, neighbor discovery
- ZDO Management
- Security Service

Zigbee Application Layer

- Application layer consists of application objects (aka end points) and Zigbee device objects (ZDOs)
- 256 End Point Addresses:
  - 240 application objects: Address EP1 through EP240
  - ZDO is EP0
  - End Points 241-254 are reserved
  - EP255 is broadcast
- Each End Point has one application profile, e.g., light on/off profile
- Zigbee forum has defined a number of profiles. Users can develop other profiles
- **Attributes**: Each profile requires a number of data items. Each data item is called an "attribute" and is assigned an 16-bit "attribute ID" by Zigbee forum

Zigbee Application Layer (Cont)

- **Clusters**: A collection of attributes and commands on them. Each cluster is represented by a 16-bit ID. Commands could be read/write requests or read/write responses
- **Cluster Library**: A collection of clusters. Zigbee forum has defined a number of cluster libraries, e.g., General cluster library contains on/off, level control, alarms, etc.
- **Binding**: Process of establishing a logical relationship (parent, child, ..)
- **ZDO**:
  - Uses device and service discovery commands to discover details about other devices.
  - Uses binding commands to bind and unbind end points.
  - Uses network management commands for network discover, route discovery, link quality indication, join/leave requests

Zigbee Application Profiles

- **Smart Energy**: Electrical, Gas, Water Meter reading
- **Commercial Building Automation**: Smoke Detectors, lights, …
- **Home Automation**: Remote control lighting, heating, doors, …
- **Personal, Home, and Hospital Care (PHHC)**: Monitor blood pressure, heart rate, …
- **Telecom Applications**: Mobile phones
- **Remote Control for Consumer Electronics**: In collaboration with Radio Frequency for Consumer Electronics (RF4CE) alliance
- **Industrial Process Monitoring and Control**: temperature, pressure, position (RFID), …
- Many others

Ref: A. Elahi and A. Gschwender, "Zigbee Wireless Sensor and Control Network," Prentice Hall, 2009, 288 pp., ISBN:0137134851,

Zigbee Home Automation (ZHA) Profile

The ZHA profile is used for smart home devices like lights, thermostats, and security sensors.

Cluster	Attribute	Description
On/Off Cluster	onOff	Controls **on/off** state of lights or switches.
Level Control	currentLevel	Adjusts **dimming** for smart bulbs.
Thermostat Cluster	localTemperature	Measures room temperature.
	occupiedHeating Setpoint	Stores heating temperature setting.
Door Lock Cluster	lockState	Stores the **locked/unlocked** state.
	pinCode	Stores a PIN for access control.
Occupancy Sensor	occupancy	Detects **motion presence** (1=occupied, 0=unoccupied).

```
Zigbee Device
 ├── Endpoint (e.g., 0x01 - Smart Light)
 │     ├── Cluster (e.g., On/Off Cluster 0x0006)
 │     │     ├── Attribute (e.g., On/Off Status, ID: 0x0000, Value:
True)
 │     │     ├── Attribute (e.g., Brightness, ID: 0x0001, Value: 180)
 │     │
 │     ├── Cluster (e.g., Level Control Cluster 0x0008)
 │           ├── Attribute (e.g., Brightness Level, ID: 0x0002,
Value: 255)
 │
 ├── Endpoint (e.g., 0x02 - Smart Plug)
 │     ├── Cluster (e.g., Power On/Off, ID: 0x0006)
```

Sample Zigbee Products

Lock
(Kwikset)

Light Bulb
(Sengled)

Hub
(Samsung)

Motion Detector
(Bosch)

Outlet
(Samsung)

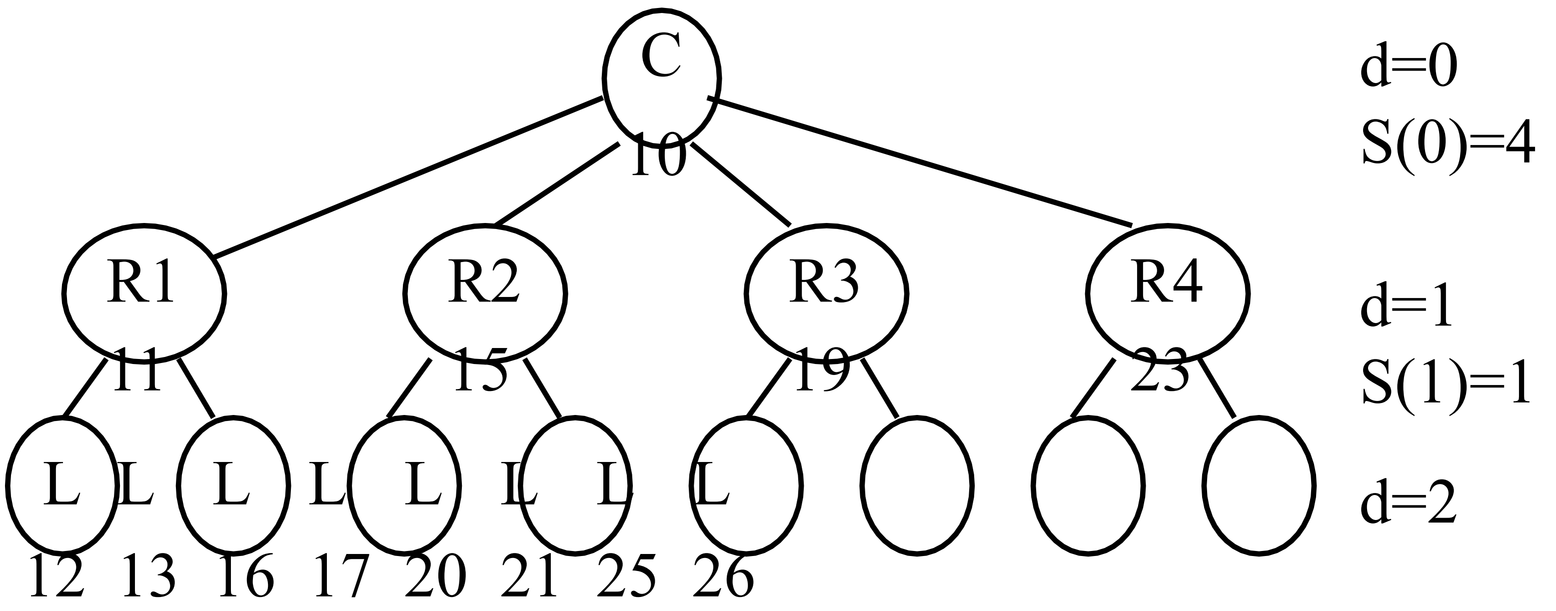
Temperature Sensor
(Visonic)

Zigbee Address Assignment

- Each node gets a unique 16-bit address
- Two Schemes: Distributed and Stochastic
- Distributed Scheme: Good for tree structure
  - Each child is allocated a sub-range of addresses.
  - Need to limit maximum depth L,
    Maximum number of children per parent C, and Maximum number of routers R: parent+1+(*n*-1)S(d)
  - Address of leaf the *n*th child is parent+(*n*-1)S(d)

$$S(d) = \begin{cases} 1 + C(L-d) & \text{if } R = 1 \\ \frac{CR^{L-d-1} - 1 - C + R}{R-1} & \text{if } R > 1 \end{cases}$$

Distributed Scheme Example



- Max depth L=2, Routers R=4, Children C=3
- Coordinator: d=0. Skip

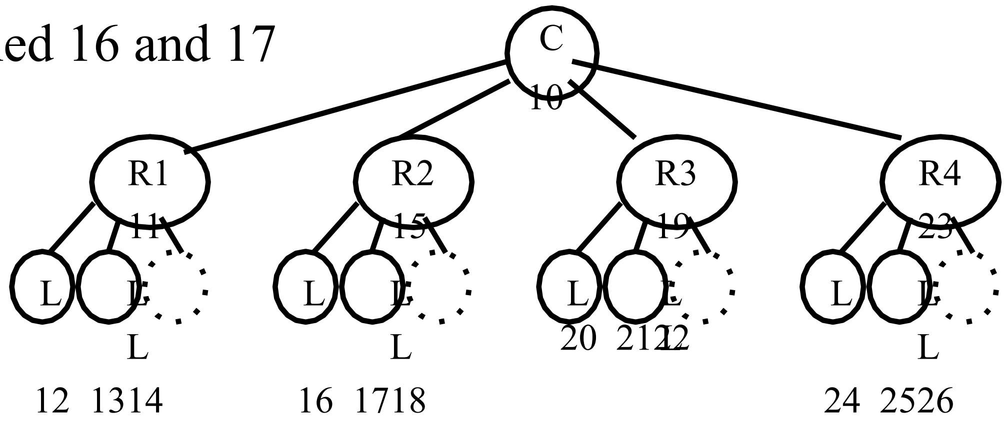
$$S(0) = \frac{CR^{L-d-1} - 1 - C + R}{R - 1} = \frac{3 \times 4^{2-0-1} - 1 - 3 + 4}{4 - 1} = 4$$

Distributed Scheme Example (Cont)

- Assume the address of coordinator is 10 (decimal)
- Address of R1 = 10+1 = 11
- Address of R2 = 10+1+S(0) = 11+4=15
- Address of R3 = 10+1+2*S(0) = 11+8 = 19
- Address of R3 = 10+1+3*S(0) = 11+12 =23
- Routers R1-R4 compute S(1):

$$S(1) = \frac{CR^{L-d-1} - 1 - C + R}{R-1} = \frac{3 \times 4^{2-1-1} - 1 - 3 + 4}{4-1} = 1$$

- Children of R1 are assigned 12 and 13
- Children of R2 are assigned 16 and 17



Stochastic Address Assignment

- Parent draws a 16 bit random number between 0 and 2^{16}-1 and assigns it to a new child. A new number is drawn if the result is all-zero (null) or all-one (broadcast). So the assigned address is between 1 and 2^{16}-2.
- Parent then advertises the number to the network
- If another node has that address an address conflict message is returned and the parent draws another number and repeats
- There is no need to pre-limit # of children or depth

Zigbee Routing

1. Ad-Hoc On-Demand Distance Vector (**AODV**)
2. Tree Hierarchical Routing
3. Many-to-one routing

Note: Zigbee does not use DSR.

AODV

- **A**d-hoc **O**n-demand **D**istance **V**ector Routing
- On-demand ⇒ Reactive ⇒ Construct a route when needed
- Avoids unnecessary computations if no traffic
- Source broadcasts Route-Request (RREQ) command to all its neighbors containing source, destination, broadcast ID
- Each node determines if this is a new request or if this copy has a lower cost. If yes, it makes a "reverse route" entry for the source in its table w previous node as the optimal reverse path.
- The node then checks if it has a route to the destination. If yes, it sends "route-reply" to the source. Otherwise, it forwards the request to all its neighbors except where it came from.
- When the source receives a "route-reply" it selects the lowest cost path and sends the packet
- If a node cannot forward the packet, it sends a "Route Error" back to the source which will re-initate route discovery.

AODV follows a **route discovery and maintenance** process:

1

Route Discovery Process
When a Zigbee device (source node) wants to send a message to another device (destination node), but **no direct route exists**, AODV initiates **route discovery**:

1. **Route Request (RREQ) Message** – Broadcasted by the source node to find a path.
2. **Route Reply (RREP) Message** – Sent back by the destination node or an intermediate node with a valid route.
3. **Route Established** – The source can now send data packets via the discovered path.

2

Route Maintenance Process
Once a route is established:

1. If an **intermediate node fails**, a **Route Error (RERR) Message** is sent back.
2. The source node then **initiates a new route discovery** to find an alternative path.

Imagine a **Zigbee Smart Home Network** with the following devices:
Smart Hub (Coordinator)
Smart Bulb (Router)
Smart Thermostat (Router)
Smart Lock (End Device)
Motion Sensor (End Device)
Scenario: The Motion Sensor Needs to Send Data to the Smart Hub, but No Direct Route Exists.

Step 1: Route Request (RREQ)
The **motion sensor** (source) sends a **Route Request (RREQ) message**.
The message is **broadcasted to all nearby nodes**.
Each node **forwards** the request **until it reaches the destination (Smart Hub)**.

Step 2: Route Reply (RREP)
The **Smart Hub** (destination) receives the RREQ and sends a **Route Reply (RREP)**.
The **reverse path** is stored in intermediate nodes.

Step 3: Data Transmission
The **motion sensor** now sends **data packets** via the **discovered route**.

Step 4: Route Maintenance
If a node **fails** (e.g., the Smart Bulb goes offline), the **Smart Thermostat detects the failure** and sends a **Route Error (RERR) message**.
The **motion sensor** then **initiates a new route discovery** to find an alternative path.

Practical Applications of AODV in Zigbee
Smart Home Automation – Ensuring reliable communication between Zigbee devices.
Industrial IoT Networks – Managing dynamic sensor networks in factories.
Smart Agriculture – Routing sensor data efficiently in large farms.

AODV Routing

- **Routing Table**: Path is not stored. Only next hop.
  - Entry = <destination, next node, "sequence #" (timestamp)>
- **Route Discovery**: Flood a **route request (RREQ)** to all neighbors. Neighbors broadcast to their neighbors

Src Addr	Req ID	Dest Addr	Src Seq #	Dest Seq #	Hop Count

- Request ID is the RREQ serial number. Used to discard duplicates.
  Source sequence # is a clock counter incremented when RREQ is sent.
  Destination sequence # is the most recent sequence from the destination that the source has seen. Zero if unknown.

Ref: K. Garg, "Mobile Computing: Theory and Practice," Pearson, 2010, ISBN: 81-3173-166-9, 232 pp., Safari Book.

AODV Routing (Cont)

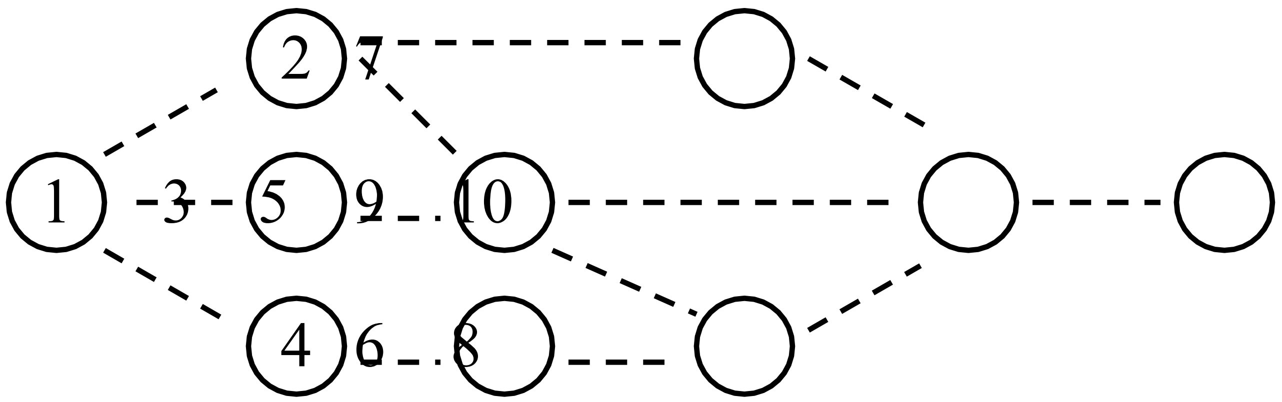
- Intermediate nodes can reply to RREQ only if they have a route to destination with higher destination sequence #
- ***Route reply* (RREP)** comes back "unicast" on the reverse path

Src Addr	Dest Addr	Dest Seq #	Hop Count	Life Time

- Destination Sequence # is from Destination's counter
  Lifetime indicates how long the route is valid
- Intermediate nodes record node from both RREP and RREQ if it has a lower cost path $\Rightarrow$ the reverse path
- Backward route to Destination is recorded if sequence number is higher <u>or</u> if sequence number is same and hops are lower
- Old entries are timed out
- AODV supports only symmetric links

AODV Routing: Example

- Node 1 broadcasts RREQ to 2, 3, 4:
  "*Any one has a route to 10 fresher than 1. This is my broadcast #1*"
- Node 2 broadcasts RREQ to 1, 5, 7
- Node 3 broadcasts RREQ to 1, 5
- Node 4 broadcasts RREQ to 1, 6



AODV Example (Con't)

Pkt #	Pkt #				Req	Src	Dest			New Table Entry			
In	Out	From	To	Message	ID	Seq #	Seq #	Hops	Action at Receipient	Dest	Seq	Hops	Next
	1	1	2	RREQ	1	1	1	1	New RREQ. Broadcast	1	1	1	1
	2	1	3	RREQ	1	1	1	1	New RREQ. Broadcast	1	1	1	1
	3	1	4	RREQ	1	1	1	1	New RREQ. Broadcast	1	1	1	1
1	4	2	1	RREQ	1	1	1	2	Duplicate Req ID. Discard				
1	5	2	7	RREQ	1	1	1	2	New RREQ. Broadcast	1	1	2	2
1	6	2	5	RREQ	1	1	1	2	New RREQ. Broadcast	1	1	2	2
2	7	3	1	RREQ	1	1	1	2	Duplicate ID. Discard				
2	8	3	5	RREQ	1	1	1	2	Duplicate ID. Discard				
3	9	4	1	RREQ	1	1	1	2	Duplicate ID. Discard				
3	10	4	6	RREQ	1	1	1	2	New RREQ. Broadcast	1	1	2	4
5	11	7	2	RREQ	1	1	1	3	Duplicate ID. Discard				
5	12	7	9	RREQ	1	1	1	3	New RREQ. Broadcast	1	1	3	7
6	13	5	3	RREQ	1	1	1	3	Duplicate ID. Discard				
6	14	5	2	RREQ	1	1	1	3	Duplicate ID. Discard				
6	15	5	9	RREQ	1	1	1	3	Duplicate ID. Discard				
6	16	5	8	RREQ	1	1	1	3	New RREQ. Broadcast	1	1	3	5
10	17	6	4	RREQ	1	1	1	3	Duplicate ID. Discard				
10	18	6	8	RREQ	1	1	1	3	Duplicate ID. Discard				
12	19	9	8	RREQ	1	1	1	4	Duplicate ID. Discard				
12	20	9	5	RREQ	1	1	1	4	Duplicate ID. Discard				
12	21	9	7	RREQ	1	1	1	4	Duplicate ID. Discard				
12	22	9	10	RREQ	1	1	1	4	New RREQ. Respond	1	1	4	9
16	23	8	6	RREQ	1	1	1	4	Duplicate ID. Discard				
16	24	8	5	RREQ	1	1	1	4	Duplicate ID. Discard				
16	25	8	9	RREQ	1	1	1	4	Duplicate ID. Discard				
22	26	10	9	RREP	1	1	6	1	New RREP. Record and forward	10	6	1	10
26	27	9	7	RREP	1	1	6	2	New RREP. Record and forward	10	6	2	9
27	28	7	2	RREP	1	1	6	3	New RREP. Record and forward	10	6	3	7
28	29	2	1	RREP	1	1	6	4	New RREP. Record and forward	10	6	4	2

Table entry at 2 for node 1

Table entry at 4 for node 1

Table entry at 9 for node 10

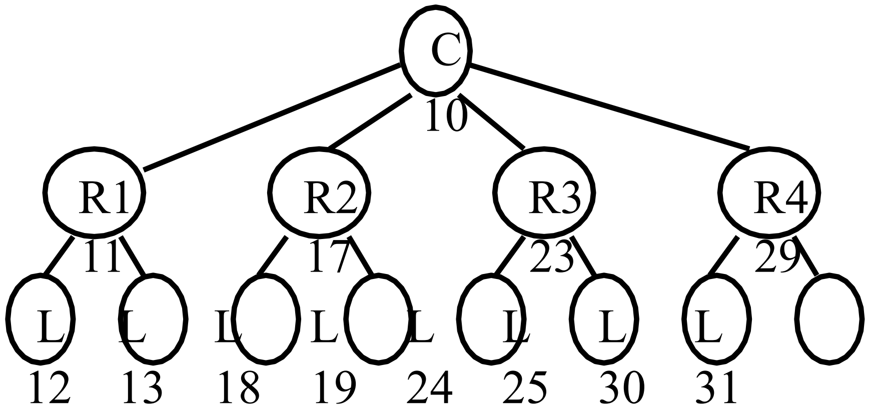
Table entry at 2 for node 10

Route Maintenance in AODV

- Each node keeps a list of active neighbors (replied to a hello within a timeout)
- If a link in a routing table breaks, all active neighbors are informed by "Route Error (RERR)" messages
- RERR is also sent if a packet transmission fails
- RERR contains the destination sequence # that failed
- When a source receives an RERR, it starts route discovery with that sequence number.
- Disadvantage: Intermediate nodes may send more up-to-date but still stale routes.
- Ref: RFC 3561, July 2003

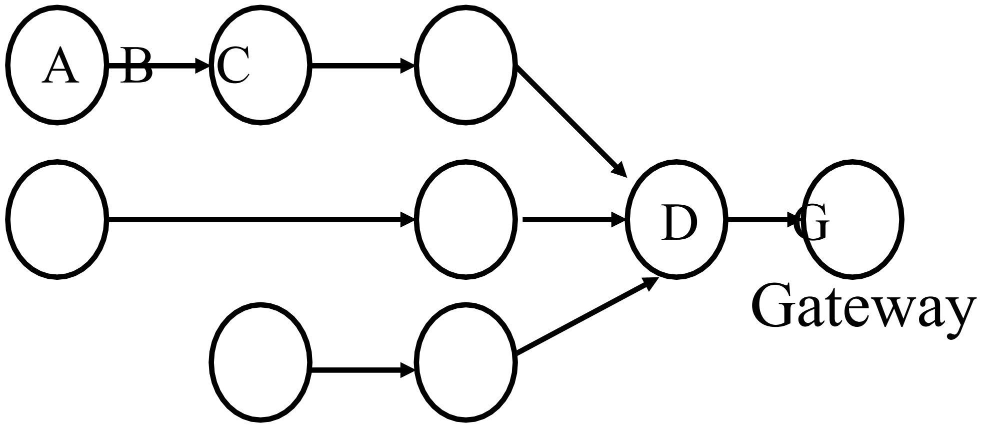
Tree Hierarchical Routing

- All leaf nodes send the packet to their parent
- Each parent checks the address to see if it is in its subrange.
  - If yes, it sends to the appropriate child.
  - If not, it sends to its parent
- Example: A12 to A30. A12 → R1 → Coordinator → R4 → A30



Many-to-One Routing

- Used for sensor data collection. All data goes to a concentrator or a gateway
- Gateway has a large memory and can hold complete routes to all nodes
- But each node only remembers the next hop towards gateway



Zigbee RF4CE

- Radio Frequency for Consumer Electronics (RF4CE) consortium developed a protocol for remote control using wireless (rather than infrared which requires line of sight)
- RF4CE merged with Zigbee and produced Zigbee RF4CE protocol
- Operates on channels 15, 20, and 25 in 2.4 GHz
- Maximum PHY payload is 127 bytes
- Two types of devices: Remotes and Targets (TVs, DVD Player,…)
- **Status Display**: Remote can show the status of the target
- **Paging**: Can locate remote control using a paging button on the target
- **Pairing**: A remote control works only with certain devices

Zigbee 2030.5

- Formerly known as "Zigbee Smart Energy 2"
- Monitor, control, automate the delivery and use of energy and water
- Adds plug-in vehicle charging, configuration, and firmware download
- Developed in collaboration with other smart grid communication technologies: HomePlug, WiFi, …
- IP based $\Rightarrow$ Incompatible with previous Zigbee

Zigbee IP

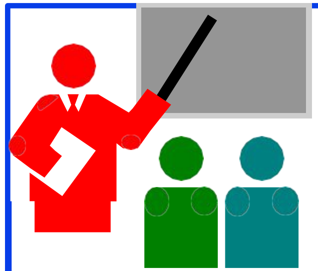
- Uses standard IPv6 frame format.
  - ⇒ Allows connecting sensors directly to Internet w/o gateways
- Uses 802.15.4 PHY, MAC and ZigBee 2030.5
- IPv6 headers are compressed using **6LowPAN**
- **RPL** Routing to discover topology
- All Internet protocols: UDP, TCP, HTTP, … can be used
- Multicast forwarding and Service discovery using multicast DNS (mDNS) and DNS Service Discovery (DNS-SD)
- Security using standard protocols: TLS (Transport Layer Security), EAP (Extensible Authentication Protocol), PANA (Protocol for carrying Authentication for Network Access)
- Not compatible with other versions of Zigbee since they use a different network layer frame format
  - ⇒ Need a gateway between Zigbee and Zigbee IP.

Ref: Zigbee Alliance, "Zigbee IP and 920IP," https://www.zigbee.org/zigbee-for-developers/network-specifications/zigbeeip/

Z-Wave

- No relationship to Zigbee but competes with it in many applications and so often confused with it
- Search for Zigbee devices on Amazon shows many products that support only Z-Wave not Zigbee
- Originally a proprietary protocol developed for remote control. Now used for IoT.
- Now standardized by Z-Wave Alliance
- Uses 915/868 MHz band
- Many IoT hubs support Z-Wave along with Zigbee

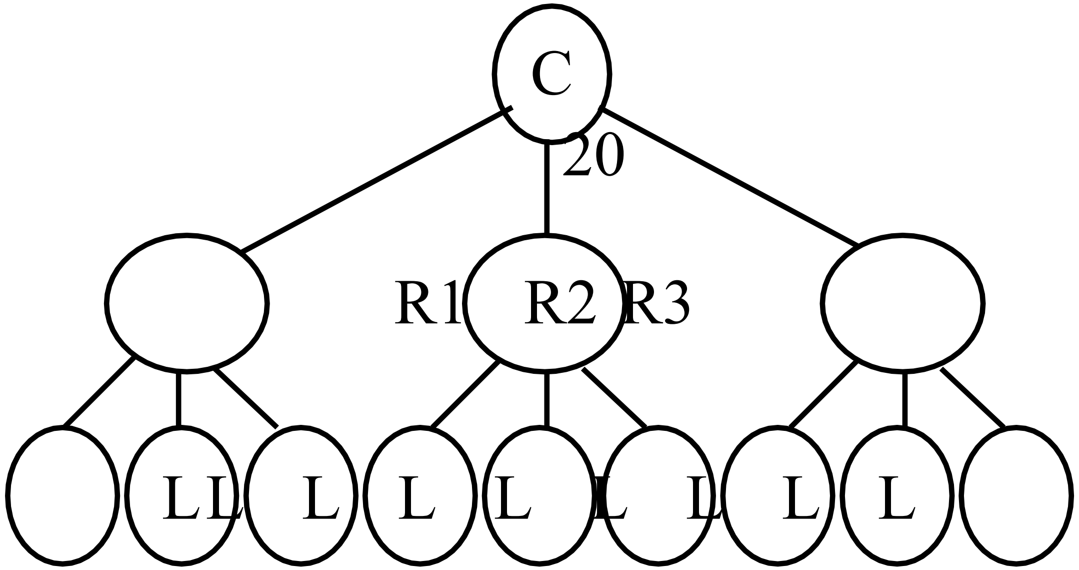
Ref: Wikipedia, “Z-Wave,” https://en.wikipedia.org/wiki/Z-Wave
Ref: Z-Wave Alliance, https://z-wavealliance.org/



Summary

1. Zigbee is an IoT protocol for sensors, industrial automation, remote control using IEEE 802.15.4 PHY and MAC
2. Zigbee PRO supports stochastic addressing, many-to-one routing, fragmentation, and mesh topologies.
3. A number of application profiles have been defined with control and management provided by ZDOs.
4. Application Support layer provides data and command communication between application objects
5. Network layer provides addressing and routing. Addressing can be assigned using distributed or stochastic schemes. Routing is via AODV, DSR, Tree Hierarchical, or many-to-one routing.
6. Zigbee RF4CE and Zigbee SEP2 are Zigbee protocols designed specifically for remote control and smart grid, respectively.

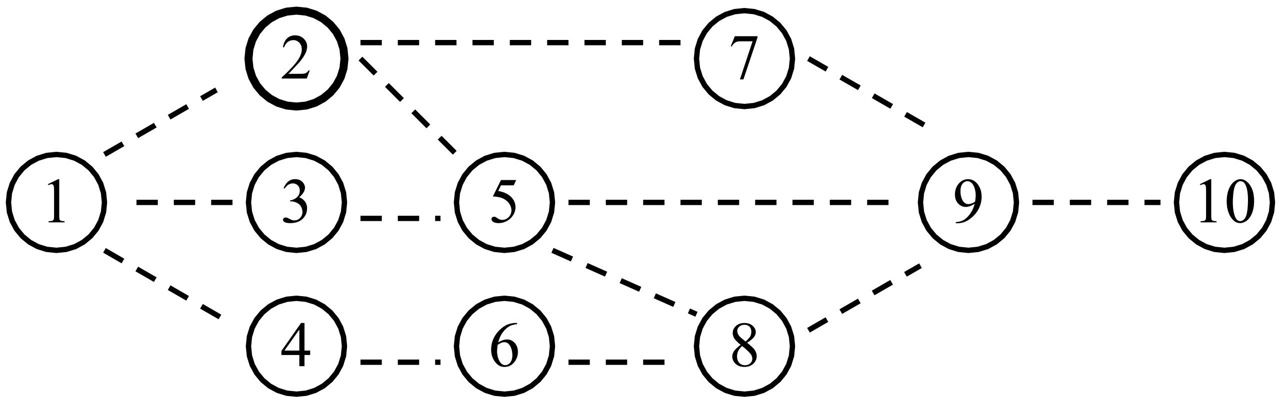
Homework 13A



- Assuming that IEEE 802.15.4 network is being planned with a maximum of 5 children per node to a depth of 2 levels and maximum 4 routers. Compute sub-ranges to be assigned to each router and the addresses assigned to each node in the network assuming the coordinator has an address of 20.

Homework 13B

- Write the sequence of messages that will be sent in the following network when node 2 tries to find the path to node 10 in the AODV example.



Reading List

- A. Elahi and A. Gschwender, “Zigbee Wireless Sensor and Control Network,” Prentice Hall, 2009, 288 pp., ISBN:0137134851, Safari Book, Chapters 2, 5, 6, 9
- K. Garg, "Mobile Computing: Theory and Practice," Pearson, 2010, ISBN: 81-3173-166-9, 232 pp., Safari Book, Sections 6.5-6.7
- R. Jain, “Networking Protocols for Internet of Things,” (6LowPAN and RPL),” http://www.cse.wustl.edu/~jain/cse570-13/m_19lpn.htm

Related Wikipedia Pages

- http://en.wikipedia.org/wiki/Zigbee
- http://en.wikipedia.org/wiki/Ad_hoc_On-Demand_Distance_Vector_Routing
- http://en.wikipedia.org/wiki/Dynamic_Source_Routing
- http://en.wikipedia.org/wiki/Source_routing
- http://en.wikipedia.org/wiki/Loose_Source_Routing

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2. O. Hersent, et al., “The Internet of Things: Key Applications and Protocols,” Wiley, 2012, 370 pp., ISBN:9781119994350, Safari Book.
3. N. Hunn, “Essentials of Short Range Wireless,” Cambridge University Press, 2010, 344 pp., ISBN:9780521760690, Safari book.
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- Zigbee Alliance, Zigbee Specification Document 053474r17, 2008
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- "How Does Zigbee Compare with Other Wireless Standards?" www.stg.com/wireless/Zigbee-comp.html

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- "Zigbee starts to buzz", Volume 50, Issue 11, Pages: 17-17, IEE Review, Nov. 2004 http://ieeexplore.ieee.org/iel5/2188/30357/01395370.pdf?tp=&arnumber=1395370&isnumber=30357
- C. Evans-Pughe,"Bzzzz zzz [Zigbee wireless standard]", Volume 49, Issue 3, Pages:28-31, IEE Review, March 2003
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Acronyms

- AODV Ad-Hoc On-Demand Distance Vector
- APS Application Support Sublayer
- APSDE Application Support Sublayer Data Entity
- APSME Application Support Sublayer Management Entity
- CSMA/CA Carrier Sense Multiple Access
- DNS Domain Name System
- DSR Dynamic Source Routing
- DVD Digital Video Disc
- EP End Point
- GHz Giga Hertz
- ID Identifier
- IEE Institution of Electrical Engineers (UK) now IET
- IEEE Institution of Electrical and Electronic Engineers
- IET Institution of Engineering and Technology
- IoT Internet of Things
- IP Internet Protocols

Acronyms (Cont)

ISM	Instrumentation, Scientific, and Medical
kB	Kilo byte
MAC	Media Access Control
MHz	Mega Hertz
NPDU	Network Protocol Data Unit
NPDU	Network Service Data Unit
PHHC	Personal, Home, and Hospital Care
PHY	Physical Layer
RF4CE	Radio Frequency for Consumer Electronics
RFC	Request for Comment
RFID	Radio Frequency ID
RREP	Route Reply
RREQ	Route Request
UWB	Ultra Wide-Band
WiFi	Wireless Fidelity

Acronyms (Cont)

- WiMAX Worldwide Interoperability for Microwave Access
- WWAN Wireless Wide Area Network
- ZDO Zigbee Device Object